Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/14/2010
PI Name:	Mulavara, Ajitkumar P. Ph.D.		
Project Title:	Development of Countermeasures to Aid Functional Egress from the Crew Exploration Vehicle Following Long Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Formerly at Universities Space Research Association		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	10/01/2009	**End Date:**	09/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Cohen, Helen (Baylor College of Medicine) Reschke, Millard (NASA Johnson Space Center) Wood, Scott (USRA/NASA JSC) Peters, Brian (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-SA02001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Crewmember adapted to the microgravity state may need to egress the vehicle within a few minutes for safety and operational reasons in various sea state conditions following a water landing. The act of emergency egress includes and is not limited to rapid motor control tasks (including both fine motor such as object manipulation and gross motor such as opening a hatch) and visual acuity tasks while maintaining spatial orientation and postural stability. Exposure to even low frequency motions (0.2-2.0 Hz) induced by sea conditions surrounding a vessel can cause significant fine and gross motor control problems affecting critical functions. These motion frequencies coupled with the varying sea state conditions (sea states 1-7 indicate frequencies ranging from 0.125-0.5 Hz, CxP 70023) can thus cause performance deficits by affecting the efficacy of motor and visual acuity dependent skills in tasks critical to emergency egress activities. Further, astronauts experience disturbances in sensorimotor function following their return to Earth due to adaptive responses that occur during exposure to the microgravity conditions of spaceflight. Thus, during exploration class missions the interaction between a debilitated crewmember during re-adaptation to Earth's gravity and the environmental constraints imposed by a water landing scenario may lead to disruption in the ability to perform functional egress tasks during the initial introduction to the Earth's gravitational environment. At present, no operational countermeasure has been implemented to mitigate this risk. Therefore, the overall goals of this project are: 1) to investigate performance of motor and visual tasks during varying sea state conditions simulated using a six degree of freedom motion platform (6 DOFMP). 2) to develop a countermeasure based on stochastic resonance that could be implemented to enhance sensorimotor capabilities with the aim of facilitating rapid adaptation to gravitational transitions following long-duration spaceflight. In the current year we are completing experiments investigating the robustness of balance performance responses to the countermeasure based on stochastic resonance. In the following year we are aiming to characterize performance on functional tasks using a 6DOFMP as a function of frequency and amplitude across different planes of motion and continue testing efficacy of the SR stimulation under otolith-canal mismatch conditions using off vertical axis rotation (OVAR) while seated in a chair. Stochastic resonance (SR) is a mechanism by which noise can assist and enhance the response of neural systems to relevant sensory signals. Application of imperceptible SR noise coupled with sensory input through the proprioceptive, visual, or vestibular sensory system has been shown to improve motor function. Specifically, studies have shown that vestibular electrical stimulation by imperceptible stochastic noise (SRVS), when applied to normal young and elderly subjects, significantly improved their ocular stabilization reflexes in response to whole-body tilt as well as balance performance during postural disturbances. The goal of the study performed in the first year of the project was to 1) study the efficacy of SRVS on improving balance performance during standing on an unstable surface; 2) optimize the characteristics of the SRVS; and 3) determining the robustness of the balance performance responses to the SRVS. Subjects performed a standardized balance task of standing on a block of 10-cm-thick medium-density foam with their eyes closed for a total of 43.5 seconds. Balance performance was measured using a force plate under the foam block and using inertial motion sensors placed on the torso and head segments. Stochastic electrical stimulation was applied to the vestibular system through electrodes placed over the mastoid process behind the ears. A new portable constant current stimulator driven by a microprocessor powered using a 3.7 V battery and capable of delivering ±5mA with subject isolation was designed and built to deliver the stimulus. Stochastic stimulation signals were generated with frequencies in the bandwidth of 1-2Hz and 0.01-30Hz. Subjects were tested at seven amplitudes of the signals, with the root mean square of the signal increasing by 30 microamperes for ±100 μA increment in the current range of 0 μA - ±700 μA. Six parameters were calculated to characterize the performance of subjects during the baseline (first half of a trial) and the stimulus (latter half of trial) periods for all seven amplitudes: RMS of - medial-lateral (ML) force; roll moment; ML linear acceleration and roll angular velocity for head and torso segments. Optimal stimulus amplitude was determined as the one at which the ratio of parameters from the stimulus period to the baseline period for any amplitude range was less than that for the no stimulus condition (0 μA) on a minimum of four of six parameters listed above. Balance performance showed significant improvement with the application of the vestibular SR stimulation in the range of 10%-25% in 10 out of 15 subjects compared with 5%-26% in 8 out of 15 subjects for the 0-30Hz and 1-2Hz frequency ranges, respectively, across all six parameters. The amplitude levels at which subjects showed the optimal improvement in performance was varied across subjects and different for the two frequency ranges even for the same subjects. Optimization of the frequency and amplitude of the stochastic noise signals for maximizing balance performance will have a significant impact on development of vestibular stochastic resonance as a unique system to aid recovery of function in astronauts after long-duration spaceflight or in people with balance disorders.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ultimately an SR based countermeasure may be fielded either as a training modality to enhance adaptability or skill acquisition, or as a miniature patch type stimulator that may be worn by people with disabilities due to aging or disease to improve posture and locomotion function. Such populations may include older adults; patients with either diabetic neuropathy or stroke.
	The goal of the project in the first year was to study the efficacy of the stochastic resonance vestibular electrical stimulation (SRVS) signals on improving balance performance during standing on an unstable surface; optimize the characteristics of the SRVS; and the robustness of the balance performance responses to the SRVS. Subjects performed a standardized balance task of standing on a block of 10-cm-thick medium-density foam with their eyes closed for a total of 43.5 seconds. Balance performance was measured using a force plate under the foam block and using inertial motion sensors placed on the torso and head segments. Stochastic electrical stimulation was applied to the vestibular system through electrodes placed over the mastoid process behind the ears. A new portable constant current stimulator driven by a ARM7 microprocessor powered using a 3.7 V battery and capable of delivering ±5mA with subject isolation was designed and built to deliver the stimulus. Stochastic stimulation signals were generated with frequencies in the bandwidth of 1-2Hz and 0.01-30Hz. Subjects were tested at seven amplitudes of the signals (0 - ± 700 μA), with the root mean square (RMS) of the signal increasing by 30 μA for each ±100 μA increment in the current range. Six parameters were calculated to characterize the balance performance of subjects during the baseline (first half of a trial) and the stimulus (latter half of trial) periods for all seven amplitudes: RMS of -medial-lateral (ML) force; roll moment; ML linear acceleration and roll angular velocity for head and torso segments. Optimal stimulus amplitude was

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Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Mulavara A, Fiedler M, Kofman I, Fisher E, Wood S, Serrador J, Peters B, Cohen H, Reschke M, Bloomberg J. "Sensorimotor challenges of space exploration - development of vestibular stochastic resonance as a countermeasure." Aerospace Medical Association 81st Annual Scientific Meeting, Phoenix, AZ, May 9-13, 2010. Aviation, Space, and Environmental Medicine 2010 Mar;81(3):216. , Mar-2010
Abstracts for Journals and Proceedings	Mulavara AP, Fiedler M, Kofman I, Fisher E, Wood S, Serrador J, Peters B, Cohen H, Reschke MF, Bloomberg JJ. "Development of countermeasures to aid functional egress from the crew exploration vehicle following long duration spaceflight." Sensorimotor Session, NASA Human Research Program Investigators' Workshop, Houston, TX February 3-5, 2010. NASA Human Research Program Investigators' Workshop, Houston, TX, Abstract Book, February 2010. , Feb-2010